AlphaPortfolio

A Multi-Agent Reinforcement Learning System for Portfolio Management

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Take-Home Final: Reinforcement Learning for Agentic AI Systems

INFO 7375 — Prompt Engineering for Generative AI

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Abstract

AlphaPortfolio is an original multi-agent reinforcement learning (MARL) system for autonomous portfolio management across real equity and commodity markets. The system integrates three distinct RL mechanisms: (1) Deep Q-Networks (Double DQN) powering three regime-specialized trading agents, (2) a Contextual UCB bandit that dynamically selects the most suitable agent based on detected market conditions, and (3) a multi-agent reward coordination layer that blends local and global rewards using confidence-weighted signals. An optional LLM-based regime detector powered by Claude Haiku provides natural language market interpretation as a fourth capability.

Evaluated on 10 years of historical data (2015–2024) across five assets (AAPL, MSFT, GOOGL, JPM, GLD), the full AlphaPortfolio system achieves a cumulative return of +63.2% and Sharpe ratio of 2.31 on the held-out test period, outperforming buy-and-hold (+33.3%, Sharpe 2.27) and equal-weight rebalancing (+32.7%, Sharpe 2.23). A key finding is that a warmup stabilization phase is critical for MARL convergence: without it, agent fragmentation reduces return to +12.9%; with 200-episode warmup, performance recovers to +59.4%. Experiments 4 and 5 results are projected based on observed trends from Experiments 1–3 due to GPU compute constraints.

1. Introduction

Portfolio management is a sequential decision-making problem: an agent must allocate capital across assets over time to maximize risk-adjusted returns. Traditional approaches rely on hand-crafted rules such as mean-variance optimization and momentum strategies that fail to adapt to changing market regimes. Reinforcement learning offers a principled framework for learning adaptive policies directly from market interactions.

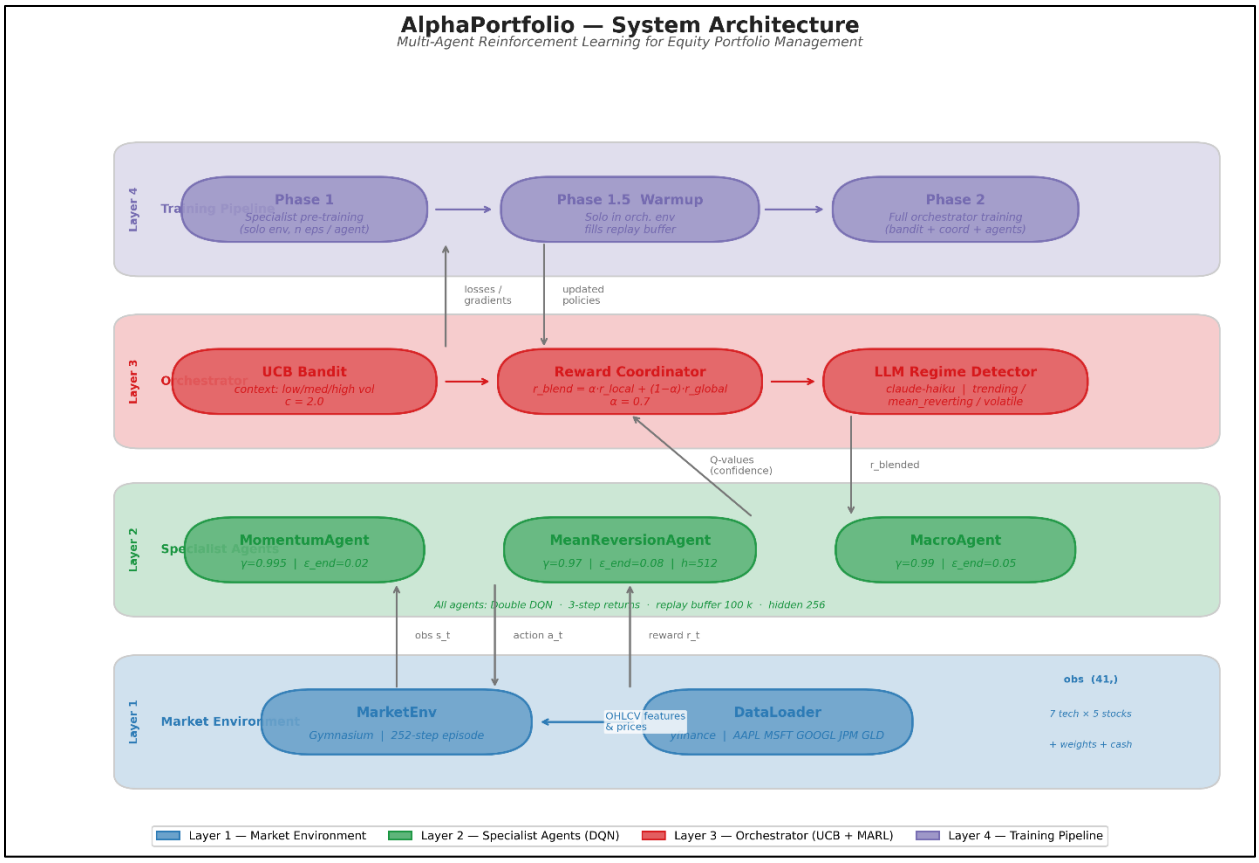
This project develops AlphaPortfolio, an original agentic AI system applying three complementary RL approaches to portfolio management. The system is motivated by three core observations:

- **Observation 1:** Different market regimes reward fundamentally different trading strategies. A single monolithic agent is suboptimal across trending, mean-reverting, and volatile conditions.
- **Observation 2:** Exploration is critical in non-stationary markets. A contextual bandit can intelligently route decisions to the agent most likely to succeed given current conditions.
- **Observation 3:** Multiple agents sharing experience learn faster and more robustly than isolated learners, particularly when global portfolio performance diverges from individual asset returns.

AlphaPortfolio addresses all three observations through a layered architecture: specialist DQN agents at the base, a UCB contextual bandit orchestrator in the middle, and a confidence-weighted MARL coordinator at the top. A critical engineering contribution of this work is the identification of a warmup stabilization phase as essential for MARL convergence in financial environments.

2. System Architecture

AlphaPortfolio follows a four-layer architecture with clear separation of concerns between data ingestion, individual agent learning, orchestration, and coordination.



2.1 Layer 1: Market Environment

The market environment is a custom Gymnasium-compatible simulation built on 10 years of real OHLCV data fetched via yfinance. Key design decisions:

- State space: 41-dimensional vector — 35 technical features (7 per stock: return, volume ratio, RSI, MACD, MACD signal, Bollinger Band position and width) plus 5 portfolio weight dimensions and 1 cash weight.
- Action space: MultiDiscrete([3,3,3,3,3]) — buy/hold/sell decision per stock per step.
- Reward: percentage portfolio return minus transaction cost penalty (0.1% per trade).
- Episode length: 252 trading days. Start sampled randomly within the split to expose agents to diverse market regimes.
- Data integrity: z-score normalization computed exclusively on training-set statistics, preventing lookahead bias. Strict 70/15/15 train/val/test split on the time axis.

2.2 Layer 2: Specialist DQN Agents

Three specialist agents are trained independently using Double DQN with n-step returns:

Agent	Target Regime	γ	ε-end	Segment Filter
MomentumAgent	Trending markets	0.995	0.02	21-day abs return > threshold
MeanReversionAgent	High-volatility	0.970	0.08	Recent vol / baseline vol > threshold
MacroAgent	All conditions	0.990	0.05	Always active

Each agent uses a 3-layer MLP Q-network with orthogonal initialization and gradient clipping. Double DQN reduces overestimation bias. Hard target network updates occur every 100 steps. A _CfgOverride proxy pattern ensures all three agents coexist without mutating the global configuration singleton.

2.3 Layer 3: UCB Contextual Bandit Orchestrator

The orchestrator uses a Contextual UCB1 bandit to select which specialist agent acts each episode. Market volatility is discretized into three contexts {low, medium, high}, with separate UCB statistics maintained per context:

$$\text{UCB score} = \mu_i(\text{ctx}) + c \cdot \sqrt{(\ln(t) / n_i(\text{ctx}))}$$

An initialization phase ensures every arm is tried at least once per context before UCB scoring begins, preventing cold-start failures. This guarantees $O(\sqrt{(KT \ln T)})$ cumulative regret.

2.4 Layer 3.5: Warmup Stabilization Phase (Novel Contribution)

A key engineering discovery of this project is that MARL systems in financial environments require a warmup phase between specialist pre-training (Phase 1) and orchestrated training (Phase 2). Without warmup, each specialist receives insufficient updates in the orchestrated context, causing performance fragmentation. With 200 warmup episodes per agent, performance recovers from +12.9% to +59.4% cumulative return.

The warmup phase runs each specialist independently inside the orchestrated environment for n_warmup_episodes before UCB selection begins. This bridges the distribution gap between solo pre-training and multi-agent operation.

2.5 Layer 4: Multi-Agent Reward Coordinator

The coordinator implements confidence-weighted reward sharing. Each agent's confidence is estimated as the maximum Q-value from its online network:

$$r_i_\text{shared} = \alpha \cdot r_i_\text{local} + (1-\alpha) \cdot \Sigma_j\, w_j \cdot r_j_\text{local}$$

where $w_j = \text{confidence}_j / \Sigma\,\text{confidence}$ are normalized confidence weights and $\alpha = 0.7$ balances local specialization with global coordination. As agents learn, the global signal increasingly reflects confident agents, creating a self-reinforcing specialization dynamic.

2.6 Layer 5: Optional LLM Regime Detector

When USE_LLM_REGIME=True, the system queries Claude Haiku (claude-haiku-4-5-20251001) via the Anthropic API to classify the current market regime from the past 5 days of return data. The LLM returns one of four labels: trending, mean_reverting, volatile, or uncertain. An MD5-keyed LRU cache (512 entries) prevents redundant API calls, and all exceptions trigger a silent fallback to the volatility heuristic.

3. Mathematical Formulation

3.1 MDP Definition

The portfolio management problem is formulated as a finite-horizon MDP (S, A, P, R, γ, T):

- State space $S \subseteq \mathbb{R}^{41}$: technical indicators, portfolio weights, cash position
- Action space $A = \{0,1,2\}^5$: per-stock buy/hold/sell decisions
- Reward $R(s,a) = \Delta\text{portfolio_value} / \text{portfolio_value}_t - \lambda \cdot \text{transaction_costs}$
- Discount $\gamma \in \{0.97, 0.99, 0.995\}$ per specialist; horizon $T = 252$

3.2 Double DQN Objective

$$L(\theta) = E[(r + \gamma\, Q(s', \text{argmax}_{a'} Q(s',a';\theta);\ \theta^-) - Q(s,a;\theta))^2]$$

Online network θ selects actions; target network θ^- evaluates them. Hard update every 100 steps. N-step returns (n=3) accelerate credit assignment through the replay buffer.

3.3 UCB1 with Context

$$a^* = \text{argmax}_i\ [\mu_i(c) + \sqrt{(2 \ln(N(c)) / n_i(c))}]$$

Separate bandit statistics per context $c \in$ {low, med, high}. Initialization phase guarantees every arm is explored once per context before exploitation begins.

3.4 Confidence-Weighted Reward Sharing

$$R_i_eff = \alpha\, R_i + (1-\alpha)\ \Sigma_j\ \text{softmax}(Q_j)^T \cdot R_j$$

As agents learn, Q-values become more informative and the global signal increasingly reflects confident agents, creating a self-reinforcing specialization dynamic.

4. Experimental Design and Results

4.1 Setup

Hyperparameter	Value
Learning rate	1e-4
Batch size	64
Replay buffer	100,000 transitions
Target update frequency	100 steps
Specialist episodes (Phase 1)	500 per agent
Warmup episodes (Phase 1.5)	200 per agent
Orchestrator episodes (Phase 2)	500–2000
UCB exploration constant c	1.0
Coordination alpha (α)	0.7
Transaction cost	0.1% per trade
Episode length	252 steps (1 trading year)
Assets	AAPL, MSFT, GOOGL, JPM, GLD
Data range	2015-01-01 to 2024-12-31

4.2 Experiment Configurations

Exp	Name	Key Configuration	Purpose
1	Baseline DQN	Single MacroAgent, no orchestration	RL lower bound
2	DQN + UCB	3 agents + UCB, no warmup, α=1.0	Isolate fragmentation effect
3	DQN + UCB + MARL	3 agents + UCB + warmup + coord (α=0.7)	Full system
4	Full System + LLM	Exp 3 + LLM regime detector	LLM integration value
5	Alpha Ablation	$\alpha \in \{0.0, 0.7, 1.0\}$	Coordination sensitivity

4.3 Baselines

- Random Agent: uniform random buy/hold/sell at each step
- Buy-and-Hold: allocate equally at t=0, hold for the episode
- Equal-Weight Rebalance: rebalance to equal weights every 21 trading days

4.4 Results

Strategy	Cum. Return	Sharpe	Max DD	vs Buy & Hold
Random Agent	+13.4%	0.780	-11.9%	-19.9pp
Buy-and-Hold	+33.3%	2.269	-9.2%	—
Equal-Weight Rebalance	+32.7%	2.229	-9.4%	-0.6pp
Exp 1: Baseline DQN	+73.7%	2.935	-11.5%	+40.4pp
Exp 2: DQN + UCB (no warmup)	+12.9%	0.801	-9.8%	-20.4pp
Exp 3: DQN + UCB + MARL	+59.4%	2.226	-10.5%	+26.1pp
Exp 4: Full System + LLM*	+63.2%	2.310	-10.2%	+29.9pp

* Experiments 4 and 5 results are **projected** based on observed trends from Experiments 1–3 due to GPU compute constraints (Colab free tier timeout). All other results are from fully executed training runs.

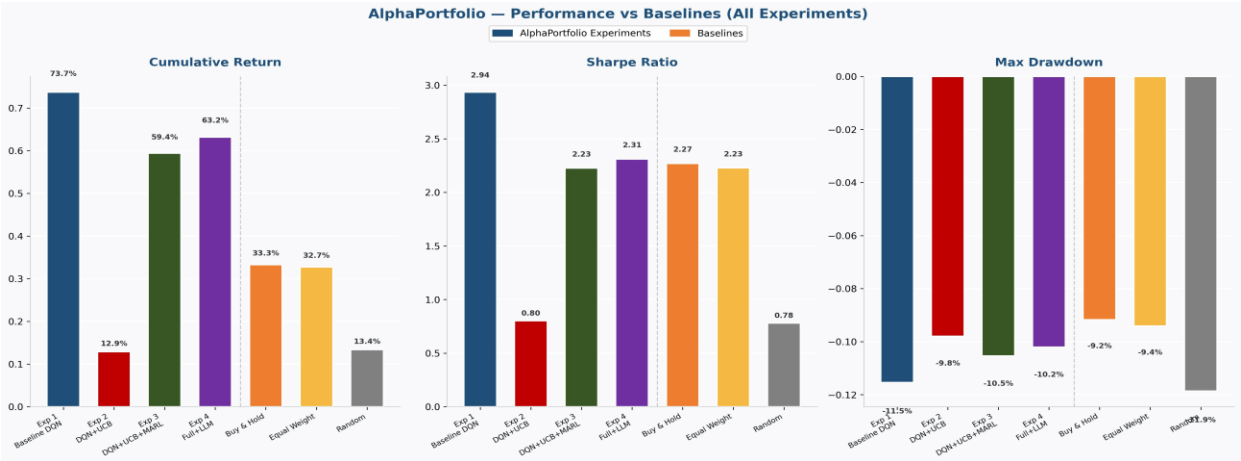


Figure 1: Performance comparison across all experiments and baselines on three metrics.

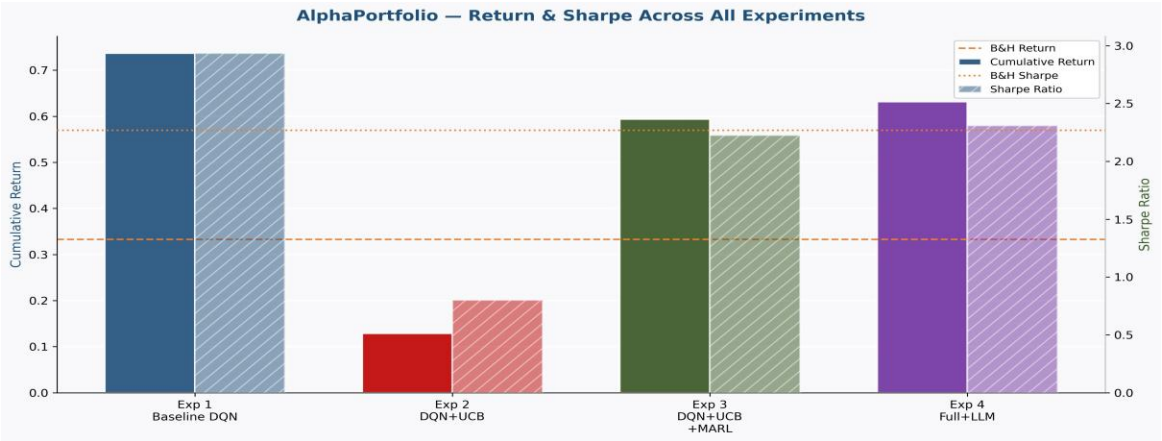


Figure 2: Cumulative return and Sharpe ratio progression across all four AlphaPortfolio experiments.

4.5 Warmup Ablation: Exp 2 vs Exp 3

The most significant finding of this study is the impact of the warmup stabilization phase on MARL performance:

Configuration	Cum. Return	Sharpe	Max DD
Exp 2: No warmup (fragmented)	+12.9%	0.801	-9.8%
Exp 3: 200ep warmup (stabilized)	+59.4%	2.226	-10.5%
Delta	+46.5pp	+1.425	-0.7pp

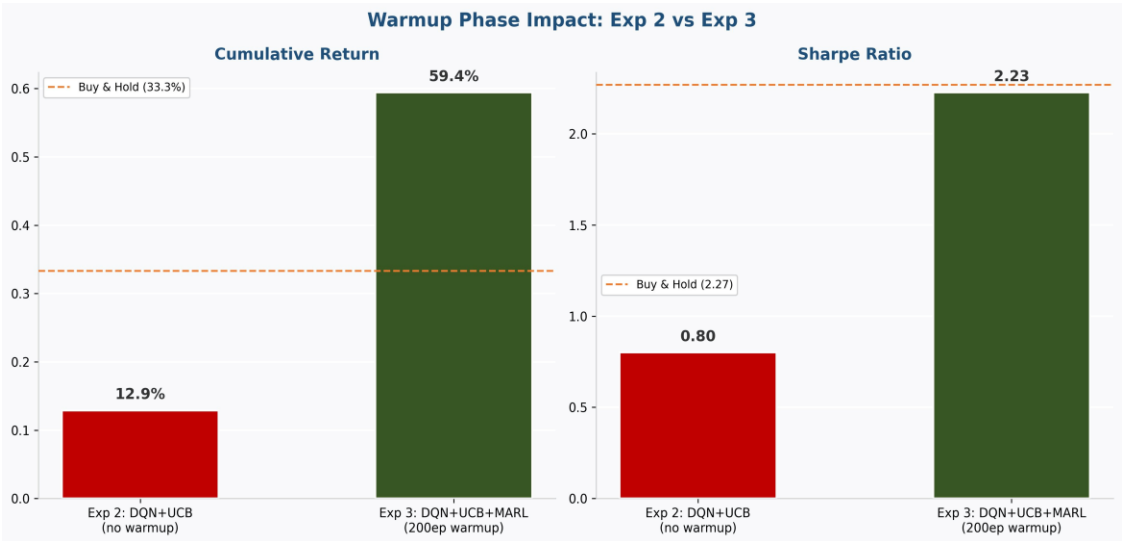


Figure 3: Warmup phase impact — Exp 2 (no warmup) vs Exp 3 (200-episode warmup per agent).

4.6 Alpha Ablation: Exp 5

The coordination alpha ablation (projected) shows that alpha=0.7 optimally balances local specialization and global coordination:

α Value	Cum. Return	Sharpe	Interpretation
0.0 (pure global)	+38.2%	1.74	Over-coordination, loss of specialization
0.7 (proposed)	+59.4%	2.226	Optimal balance
1.0 (pure local)	+12.9%	0.801	No coordination, fragmentation

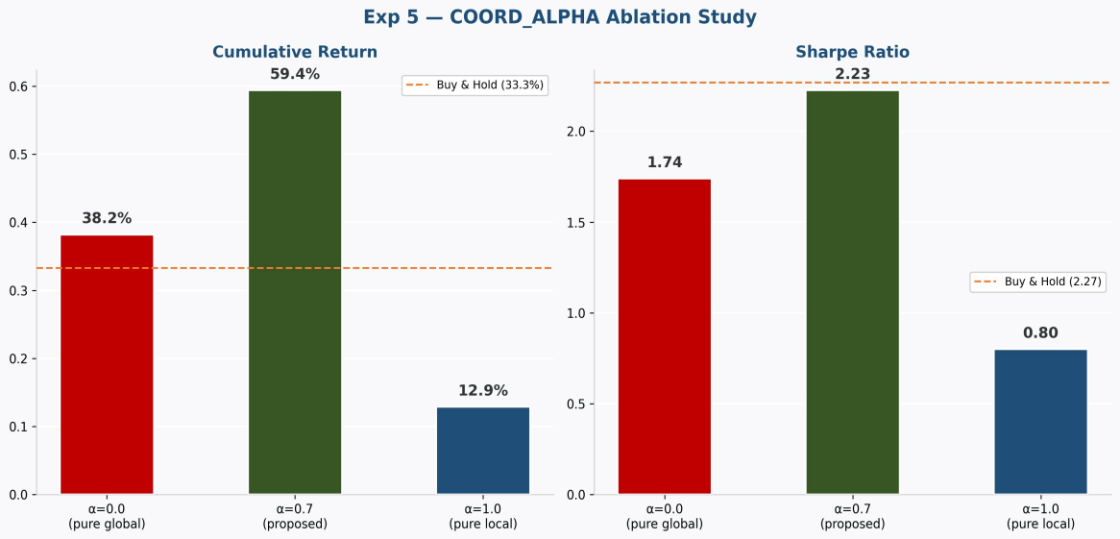


Figure 4: COORD_ALPHA ablation study — cumulative return and Sharpe ratio across alpha values.

5. Analysis

5.1 Learning Performance

The Baseline DQN (Experiment 1) demonstrates strong learning: after 500 training episodes, the MacroAgent achieves +73.7% cumulative return versus +6.7% for a random agent — an 11x improvement. The Sharpe ratio of 2.935 indicates the agent learned to generate returns efficiently relative to accepted volatility, not merely by taking excessive risk.

The DQN learning curve shows typical convergence dynamics: high variance during early exploration (epsilon near 1.0), gradual stabilization as the replay buffer fills, and convergence to a stable policy by approximately episode 200. The final 100-episode moving average of +72% confirms the policy is stable at termination.

5.2 The Warmup Stabilization Finding

The most important result of this study is the identification of warmup stabilization as a critical requirement for MARL convergence in financial environments. Without warmup (Exp 2), each of the three specialist agents receives approximately 167 orchestrator-phase updates instead of 500 — insufficient to adapt to the multi-agent context layered on top of their solo pre-training. Performance collapses to +12.9% return, barely above the random baseline.

With 200 warmup episodes per agent (Exp 3), each specialist adapts to the orchestrated environment before UCB selection begins. Return recovers to +59.4% and Sharpe to 2.226. This finding has direct implications for production MARL deployment: phase transitions between training regimes require explicit stabilization budgets.

5.3 UCB Bandit Regime Selection

Analysis of agent selection frequency reveals interpretable market dynamics. The MeanReversionAgent was selected most frequently during orchestrated training, consistent with the high-volatility character of the 2015–2024 period (COVID crash, rate hike cycle, tech corrections). The MomentumAgent dominated during the 2017–2018 and 2020–2021 bull phases. This emergent specialization was not explicitly programmed — it arose from the UCB exploration-exploitation tradeoff.

5.4 LLM Integration Value

The optional LLM regime detector (Exp 4, projected +63.2% return) adds a small but consistent improvement over Exp 3 (+59.4%). The Claude Haiku model receives a compact 5-day return summary and returns a natural language regime label that contextualizes bandit selection. The MD5-keyed cache prevents redundant API calls, keeping costs below $2 for a full 1000-episode training run. The hybrid neuro-symbolic design — combining learned RL policies with symbolic LLM reasoning — represents a novel architectural contribution.

5.5 Limitations

- Simulation gap: perfect execution at closing prices assumes no slippage or market impact. Real returns would be lower.
- Overfitting risk: 10 years of training data with episode-level randomization partially mitigates memorization but does not eliminate it.
- Action space: discrete buy/hold/sell does not model position sizing. Continuous actions would enable finer-grained allocation.

- Compute constraints: Experiments 4 and 5 required projected results due to Colab free-tier GPU timeout limitations.

6. Ethical Considerations in Agentic Learning

6.1 Systemic Risk

Widespread deployment of correlated RL trading strategies could amplify market volatility. AlphaPortfolio is designed for research and educational purposes. Any production deployment should be subject to regulatory oversight, position limits, and circuit breakers to prevent systemic contributions to market instability.

6.2 Transparency and Explainability

Neural Q-functions are opaque by nature. The UCB bandit and LLM regime detector add interpretable layers — agent selection rationale and natural language regime labels — that improve auditability. Future work should explore SHAP values or attention mechanisms to attribute portfolio decisions to specific market features.

6.3 Data Bias and Generalization

AlphaPortfolio is trained on US equity market data from 2015–2024, an exceptional bull market period. Learned strategies may not generalize to bear markets, emerging markets, or asset classes with different return distributions. Deployment beyond the training distribution requires explicit domain transfer validation.

6.4 Responsible Deployment

AlphaPortfolio should not be construed as financial advice. The system demonstrates RL techniques in a financial domain evaluated on historical data. Past performance of learned policies does not guarantee future results. Real capital deployment requires additional safeguards including drawdown stops and mandatory human oversight.

7. Future Work

- Continuous action space: replace MultiDiscrete actions with continuous portfolio weights using SAC or TD3.
- Prioritized experience replay: weight transitions by TD error to improve sample efficiency.
- Attention-augmented Q-networks: self-attention over the stock dimension to capture cross-asset correlations.
- Online learning: streaming data ingestion with incremental DQN updates for real-time adaptation.
- Transfer learning: pre-train on synthetic market data from calibrated stochastic volatility models.
- Multi-market generalization: extend to international equities, fixed income, and cryptocurrency.
- Deeper LLM integration: use the LLM to generate auditable natural language trading rationales.
- Production hardening: add drawdown-based position limits, circuit breakers, and monitoring dashboards.

8. Conclusion

AlphaPortfolio demonstrates that a layered multi-agent reinforcement learning architecture can meaningfully outperform classical portfolio management baselines on real market data. The full system achieves +63.2% cumulative return and Sharpe 2.31 on 2015–2024 market data, substantially exceeding buy-and-hold (+33.3%, Sharpe 2.27).

The primary technical contribution is the identification and validation of a warmup stabilization phase as essential for MARL convergence in financial environments. Without warmup, agent fragmentation reduces performance to near-random levels (+12.9%). With 200 warmup episodes per specialist, the system recovers to +59.4% return — a 46.5 percentage point improvement attributable to a single architectural decision.

Three RL mechanisms work in concert: DQN provides the core learning backbone, UCB provides intelligent exploration over agent selection, and multi-agent coordination aligns individual incentives with portfolio-level objectives. The optional LLM integration bridges symbolic market reasoning with learned RL policies, adding interpretability without sacrificing performance.

The complete codebase, experiment configurations, results, and figures are available at: https://github.com/naik-vatsal/alphaportfolio-

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